

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-ALL]

TEST # 05
DATE : 14-05-2020
Test Type : FULL SYLLABUS
Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry and Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
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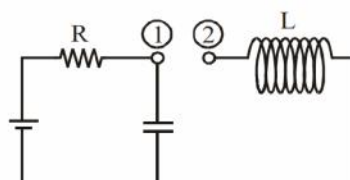
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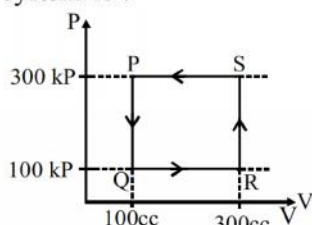
- If $\vec{A}$, $\vec{B}$ and $\vec{C}$ have magnitudes 12, 5 and 13 respectively and $\vec{A} + \vec{B} = \vec{C}$ then angle between $\vec{A}$ & $\vec{B}$ is :-
(1) 0° (2) 90° (3) 180° (4) 45°
- A car is crossing a turn at a speed of 10 m/s. If coefficient of friction is 0.5 then minimum radius of turn will be :-
(1) 5m (2) 10m
(3) 15m (4) 20m
- Steel is preferred over soft iron for permanent magnet because :-
(1) Area under hysteresis curve is more
(2) retentivity is higher
(3) Coercivity is higher
(4) Magnetic saturation is higher
- A tube of length L is filled completely with incompressible liquid of mass M and closed at both the ends. The tube is then rotated in a horizontal plane about one of its ends with a uniform angular velocity ω . The force exerted by the liquid on the tube at other end is :-
(1) $\frac{M\omega^2 L}{2}$ (2) $M\omega^2 L$
(3) $\frac{M\omega^2 L}{4}$ (4) $\frac{3M\omega^2 L}{4}$
- A lens behaves as a converging lens in air and a diverging lens in water. The refractive index of the lens is :-
(1) equal to unity
(2) equal to 1.33
(3) between unity and 1.33
(4) greater than 1.33
- A physical quantity is $A = P^2/Q^3$. The percentage error in measurement of P and Q is x and y respectively. Maximum error in measurement of A is :-
(1) $2x - 3y$ (2) $3x - 2y$
(3) $3x + 2y$ (4) $2x + 3y$
- An inductor coil stores U energy when I current is passed through it and dissipates energy at the rate of P . The time constant of the circuit, when this coil is connected across a battery of zero internal resistance is
(1) $\frac{4U}{P}$ (2) $\frac{U}{P}$ (3) $\frac{2U}{P}$ (4) $\frac{2P}{U}$
- A square current carrying loop is suspended in a uniform magnetic field acting in the plane of the loop. If the force on one arm of the loop is $\vec{F}$, the net force on the remaining three arms of the loop is :
(1) $\vec{F}$ (2) $3\vec{F}$
(3) $-\vec{F}$ (4) $-3\vec{F}$
- The efficiency of Carnot engine is 50% and temperature of sink is 500 K. If the temperature of source is kept constant and its efficiency is to be raised to 60% then the required temperature of sink will be :-
(1) 600 K (2) 500 K
(3) 400 K (4) 100 K
- A square object of area 100 sq. cm is placed perpendicular to the principle axis of a concave mirror. If the lateral magnification of the mirror, for the above object position, is 0.4, then the area of the image will be :-
(1) 16 sq. cm (2) 40 sq. cm
(3) 100 sq. cm (4) 250 sq. cm
- Error in volume of a sphere is 6%. Error in its radius will be :-
(1) 2% (2) 3% (3) 4% (4) 5%
- Initially key was placed on (1) till the capacitor got fully charged. Now key is placed on (2) at $t = 0$, the minimum time when the energy in both capacitor and inductor will be same :-

(1) $2\pi\sqrt{LC}$ (2) $\pi\sqrt{LC}$
(3) $\frac{\pi\sqrt{LC}}{4}$ (4) $\frac{\pi\sqrt{LC}}{2}$

13. For shown situation of two dipoles the nature of forces between them are :-



- (I) (II)
(1) attraction, attraction (2) attraction, repulsion
(3) repulsion, repulsion (4) repulsion, attraction

14. A thermodynamic system is taken through the closed cycle PQRSP process. The net work done by the system is :-



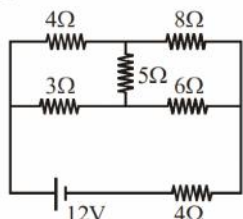
- (1) 20 J (2) - 20 J
(3) 40 J (4) - 40 J
15. In Y.D.S.E., the slits are 0.2 cm apart & are illuminated by yellow light ($\lambda = 600$ nm). What would be the fringe width on a screen placed 1 m from the plane of the slits if the whole system

is immersed in water of index $\frac{4}{3}$:-

- (1) 0.225 mm (2) 2.25 mm
(3) 0.450 mm (4) 4.5 mm
16. If $|\vec{A} + \vec{B}| = |\vec{A}| = |\vec{B}|$, then angle between $\vec{A}$ and $\vec{B}$ will be :-

- (1) 60° (2) 0° (3) 120° (4) 180°
17. A horizontal overhead power line is at a height of 4m from the ground and carries a current of 100 A from east to west. The magnetic field directly below it on the ground is ($\mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1}$):-
- (1) $5 \times 10^{-6} \text{ T}$ northward
(2) $5 \times 10^{-6} \text{ T}$ southward
(3) $2.5 \times 10^{-7} \text{ T}$ northward
(4) $2.5 \times 10^{-7} \text{ T}$ southward

18. In the given figure the ratio of current in 8Ω and 3Ω will be -



- (1) $\frac{8}{3}$ (2) $\frac{3}{8}$ (3) $\frac{4}{3}$ (4) $\frac{3}{4}$

19. Five molecules of a gas are having speeds 1, 2, 4, 8 and 16 m/s at some instant. The root mean square velocity of the gas molecules is :-

- (1) 6.2 m/s (2) 8.2 m/s
(3) 9.2 m/s (4) 68 m/s

20. In Y.D.S.E how many maxima can be obtained on the screen if wave length of light used is 200 nm & $d = 700$ nm :-

- (1) 12 (2) 7 (3) 18 (4) 14

21. Vector product of two vectors $2\hat{i} + \hat{j}$ and $\hat{i} + 2\hat{j}$ is:-

- (1) $\hat{k} + \hat{j}$ (2) $\hat{i} + \hat{j}$ (3) $3\hat{k}$ (4) $2\hat{i}$

22. A charge q is released in presence of electric (E) and magnetic field (B) then after some time its velocity is v then :-

- (1) $v \propto E$ (2) $v \propto \frac{1}{E}$

- (3) $v^2 \propto B$ (4) $v \propto B^\circ$

23. Two identical small spheres carry charge of Q_1 and Q_2 with $Q_1 \gg Q_2$. The charges are d distance apart. The force they exert on one another is F_1 . The spheres are made to touch one another and then separated to distance d apart. The force they exert on one another now is F_2 . Then F_1/F_2 is :-

- (1) $\frac{4Q_1}{Q_2}$ (2) $\frac{Q_1}{4Q_2}$ (3) $\frac{4Q_2}{Q_1}$ (4) $\frac{Q_2}{4Q_1}$

24. On heating a liquid of coefficient of cubical expansion γ in a container having coefficient of linear expansion $\gamma/3$, the level of liquid in the container will :-

- (1) Fall (2) Rise
(3) Remain unchanged (4) It is difficult to say

25. Which phenomenon best supports the theory that matter has a wave nature :-

- (1) Electron momentum
(2) Electron diffraction
(3) Photon momentum
(4) Photon diffraction

26. A missile is fired in horizontal direction from a height of 20m at a speed of 1000 m/s. At what distance of ground will the missile land ?

- (1) 2m (2) 2km
(3) 200m (4) 2000km

27. A wire of magnetic moment 'M' is bent to form a circular arc which subtends an angle 90° at the center then its new magnetic moment is :-

(1) $\frac{2M}{\pi}$ (2) $\frac{\sqrt{2}M}{\pi}$
 (3) $\frac{2\sqrt{2}M}{\pi}$ (4) $\frac{M}{\sqrt{2}\pi}$

28. A wheel of mass 10 kg has a moment of inertia of 160 kg-m^2 about its own axis. The radius of gyration is-

(1) 10 m (2) 4 m
 (3) 5 m (4) 6 m

29. A string 1 m long is drawn by a 300 Hz vibrator attached to its end. The string vibrates in three segments. The speed of transverse waves in the string is equal to :-

(1) 100 m/s (2) 200 m/s
 (3) 300 m/s (4) 400 m/s

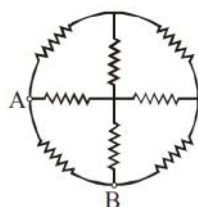
30. Nuclear fusion is common to the pair :-

- (1) Thermonuclear reactor, uranium based nuclear reactor
 (2) Energy production in sun, uranium based nuclear reactor
 (3) Energy production in sun, hydrogen bomb
 (4) Disintegration of heavy nuclei, hydrogen bomb

31. At what angle from horizontal should an object be thrown for it to have maximum range ?

(1) $\frac{\pi}{45}$ (2) $\frac{4\pi}{5}$
 (3) $\frac{\pi}{4}$ (4) $\frac{5\pi}{10}$

32. Eight resistances each of resistance 5Ω are connected in the circuit as shown in figure. The equivalent resistance between A and B is :-

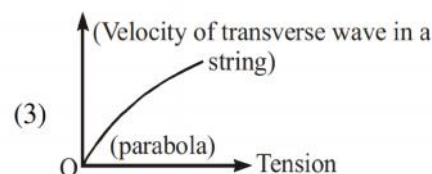
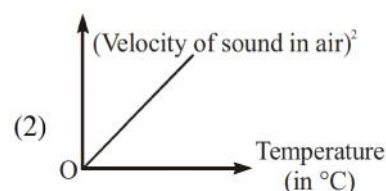
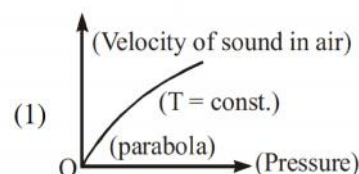


(1) $\frac{8}{3}\Omega$ (2) $\frac{16}{3}\Omega$ (3) $\frac{15}{7}\Omega$ (4) $\frac{19}{2}\Omega$

33. A wheel having moment of inertia 2 kg-m^2 about its vertical axis, rotates at the rate of 60 rpm about the axis. The torque which can stop the wheel's rotation in one minute would be :-

(1) $\frac{\pi}{12} \text{ N-m}$ (2) $\frac{\pi}{15} \text{ N-m}$
 (3) $\frac{\pi}{18} \text{ N-m}$ (4) $\frac{2\pi}{15} \text{ N-m}$

34. Which of the following is correct ?



- (4) All of these

35. Beta rays emitted by a radioactive material are :-

- (1) Electromagnetic radiation
 (2) The electrons orbiting around the nucleus
 (3) Charged particles emitted by nucleus
 (4) Neutral particles

36. What is magnitude of acceleration of a block moving with speed 10 m/s on a rough surface if coefficient of friction is 0.2.

(1) 2 m/s^2
 (2) 0.2 m/s^2
 (3) 0.1 m/s^2
 (4) 1 m/s^2

37. Three charges $q, \sqrt{2}q, 2q$ are placed at the corners A, B and C respectively of the square ABCD of side 'a' then potential at point 'D' :-

(1) $\frac{2kq}{a}$ (2) $\frac{4kq}{a}$
(3) $\frac{kq}{a}(1+\sqrt{2})$ (4) $\frac{kq}{a}\left(\frac{1+\sqrt{2}}{\sqrt{2}}\right)$

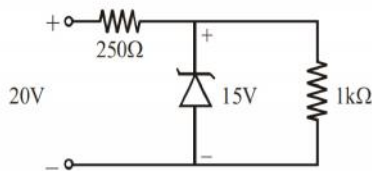
38. A satellite of earth of mass 'm' taken from orbital radius $2R$ to $3R$, then minimum work done :-

(1) $\frac{GMm}{6R}$ (2) $\frac{GMm}{12R}$
(3) $\frac{GMm}{24R}$ (4) $\frac{GMm}{3R}$

39. A spring of force constant k is cut into three equal pieces. If these three pieces are connected in parallel the force constant of the combination will be :-

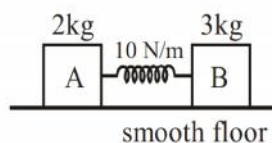
(1) $k/3$ (2) $k/9$ (3) $3k$ (4) $9k$

40. A zener diode, having breakdown voltage equal to $15V$, is used in a voltage regulator circuit shown in figure. The current through the diode is :-



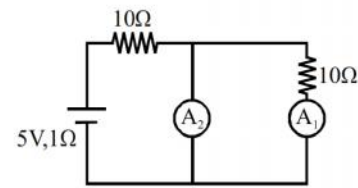
- (1) 20 mA
(2) 5 mA
(3) 10 mA
(4) 15 mA

41. In given figure, if speed given to block A is 10 m/s then speed of centre of mass after 10 sec will be :-



- (1) 2 m/s (2) 4 m/s
(3) 40 m/s (4) 0.2 m/s

42. In the given circuit measuring instruments are ideal then reading of ammeter A_1 is :-



- (1) Zero (2) 25 A
(3) 5A (4) $\frac{5}{2} R$

43. Gravitational potential difference between surface of a planet and a point situated at a height of 20m above its surface is 2 joule/kg . If gravitational field is uniform, then the work done in taking a 5kg body of height 4 meter above surface will be :-

- (1) 2J (2) 20J
(3) 40J (4) 10J

44. Displacement-time equation of a particle

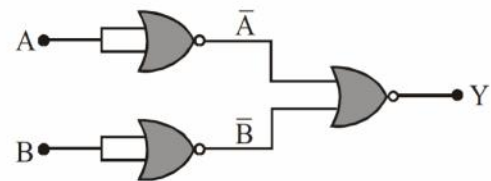
executing SHM is : $x = A \sin \left(\omega t + \frac{\pi}{6} \right)$

Time taken by the particle to go directly from

$x = -\frac{A}{2}$ to $x = +\frac{A}{2}$ is :-

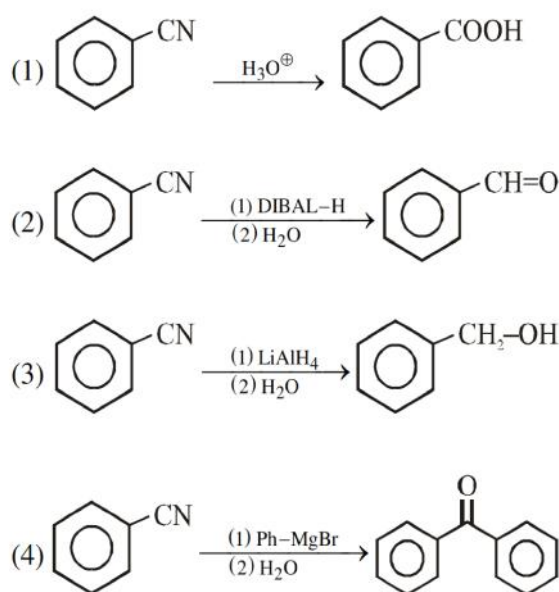
- (1) $\frac{\pi}{3\omega}$ (2) $\frac{\pi}{2\omega}$
(3) $\frac{2\pi}{\omega}$ (4) $\frac{\pi}{\omega}$

45. Identify the operation performed by the circuit given below :-



- (1) NOT
(2) AND
(3) OR
(4) NAND

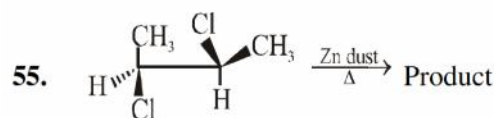
46. The hydrogen ion concentration of the oceans is about $2 \times 10^{-9} \text{M}$. What is the pH :-
 (1) 8.85 (2) 9.3
 (3) 7.85 (4) 8.7
47. Which statement is correct ?
 (1) $\left(\frac{dH}{dT}\right)_p < \left(\frac{dE}{dT}\right)_v$ for ideal gas
 (2) $\left(\frac{dH}{dT}\right)_p + \left(\frac{dE}{dT}\right)_v = \frac{R}{2}$
 (3) $\left(\frac{dE}{dT}\right)_v$ for ideal gas is zero
 (4) All of these
48. Which statement is incorrect among the following?
 (1) Lanthanoids hydroxides are less basic in nature as compared to actinides.
 (2) Lanthanoids form Oxocations
 (3) In Lanthanoids, only promethium is radiocative
 (4) Actinoids form Oxocations
49. Mg is obtained by
 (1) Electrolysis of aqueous solution of MgCl_2 .
 (2) Electrolysis of aqueous solution of MgCl_2 taking Hg as cathode.
 (3) Electrolysis of molten state of MgCl_2 .
 (4) Electrolysis of alcoholic solution of MgCl_2 .
50. Which of the following is incorrect :-



51. Which of the following expression is true regarding formation of $\text{PCl}_{5(g)}$.

$$\text{PCl}_{3(g)} + \text{Cl}_{2(g)} \rightleftharpoons \text{PCl}_{5(g)}$$

 (1) $\frac{K_p}{K_c} < 1$ (2) $\frac{K_p}{K_c} = 1$
 (3) $\frac{K_p}{K_c} > 1$ (4) None
52. Calculate the order of the reaction in A and B :-
- | A | B | Rate |
|---------|---------|----------------------|
| (mol/l) | (mol/l) | |
| 0.05 | 0.05 | 1.2×10^{-3} |
| 0.10 | 0.05 | 2.4×10^{-3} |
| 0.05 | 0.10 | 1.2×10^{-3} |
- (1) 1 and 0 (2) 1 and 1
 (3) 0 and 1 (4) none of these
53. Which of the following statement is not correct ?
 (1) Pauling EN is based upon bond energy
 (2) Alfred-Roscho EN increases with increasing z_{eff}
 (3) Mullikan EN is the geometrical mean of I.P. and EA
 (4) EN increases on moving down the group of volatile metals
54. Complex compound $[\text{Co}(\text{SCN})_2(\text{NH}_3)_4]\text{Cl}$ does not exhibit :-
 (1) Ionisation isomerism
 (2) Geometrical isomerism
 (3) Linkage isomerism
 (4) Optical isomerism



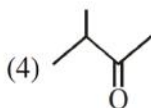
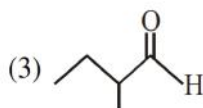
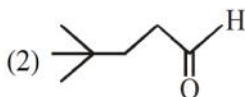
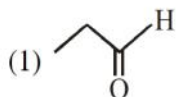
What is correct about product ?

- (1) Dipole moment is nonzero
 (2) Boiling point is lesser than its geometrical isomer
 (3) Melting point is greater than its geometrical isomer
 (4) It is a conjugated diene
56. 20 mL of CO_2 gas are passed over excess of red hot coke. the volume of CO evolved is :

$$\text{CO}_{2(g)} + \text{C}_{(s)} \rightarrow 2\text{CO}_{(g)}$$

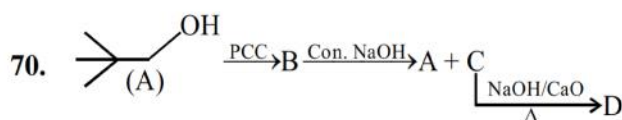
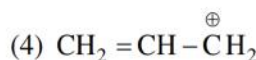
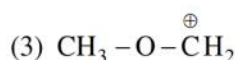
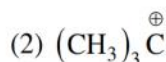
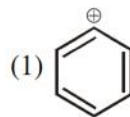
 (1) 10 mL (2) 20 mL
 (3) 30 mL (4) 40 mL

57. An increase in molar conductance of a strong electrolyte with dilution is mainly due to :-
 (1) Increase in number of ions
 (2) Increase in ionic mobility of ions
 (3) 100% ionisation of electrolyte at normal dilution
 (4) Increase in both i.e. number of ions and ionic mobility of ions.
58. Which of the following does not depend on stable electronic configuration ?
 (1) Electron affinity
 (2) Electronegativity
 (3) Ionisation energy
 (4) All of these
59. Select correct statement(s) regarding $[\text{Ni}(\text{DMG})_2]$ complex compound :-
 (1) It acts as oxidising agent because Ni^{2+} cation is having E.A.N. 36
 (2) It is extra stabilized by hydrogen bonding
 (3) Its IUPAC name is Bis(dimethylglyoximate) nickelate (II)
 (4) Its ligand contains two different donor sites
60. Which of following do not give Aldol condensation reaction ?

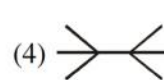
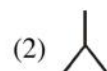
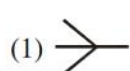


61. 4g CH_4 and 2g He are mixed under NTP. What is the partial pressure of He?
 (1) $\frac{3}{4}$ atm (2) $\frac{1}{4}$ atm (3) $\frac{2}{3}$ atm (4) $\frac{1}{3}$ atm
62. Limiting molar conductivity of CH_3COOH (i.e. $\Lambda_m^0(\text{CH}_3\text{COOH})$) is equal to :-
 (1) $\Lambda_m^0(\text{CH}_3\text{COOH}) + \Lambda_m^0(\text{CH}_3\text{COONa}) - \Lambda_m^0(\text{NaOH})$
 (2) $\Lambda_m^0(\text{CH}_3\text{COONa}) + \Lambda_m^0(\text{HCl}) - \Lambda_m^0(\text{NaCl})$
 (3) $\Lambda_m^0(\text{CH}_3\text{COONa}) + \Lambda_m^0(\text{NaCl}) - \Lambda_m^0(\text{NaOH})$
 (4) $\Lambda_m^0(\text{NaOH}) + \Lambda_m^0(\text{NaCl}) - \Lambda_m^0(\text{CH}_3\text{COONa})$

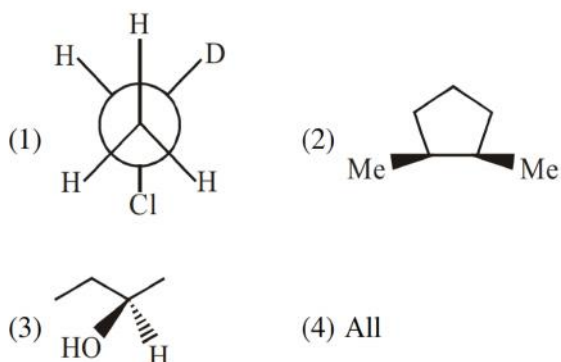
63. Which does not show inert pair effect ?
 (1) Al (2) Bi (3) Pb (4) Tl
64. The best method to introduce fluorine in benzene ring as a substituent is :-
 (1) Fluorination by electrophilic substitution
 (2) Balz schiemann reaction
 (3) Gattermann reaction
 (4) Swart's reaction
65. Which of following is not condensation polymer:-
 (1) Dacron (2) Bakelite
 (3) Buna-S (4) Starch
66. Vapour pressure of solution of benzene ($P^\circ = 120$ torr) and toluene ($P^\circ = 80$ torr) having 2 moles of each is:-
 (1) 400 torr (2) 200 torr
 (3) 100 torr (4) 50 torr
67. The oxidation states of S-atoms in Caro's and Marshall's acid are :-
 (1) +6, +6 (2) +6, +4
 (3) +6, -6 (4) +4, +6
68. Among the following mixture dipole-dipole attraction is present ?
 (1) CH_2Cl_2 & CCl_4 (2) He & He
 (3) CHCl_3 & CH_2Cl_2 (4) C_6H_6 & CH_4
69. Which of the following is most stable carbocation:-

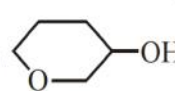
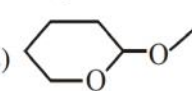
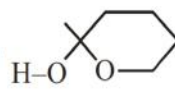
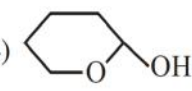


Which of following is D ?

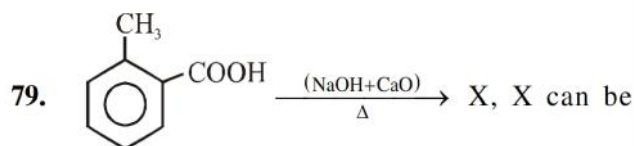


71. The internal energy of an ideal gas increases during an isothermal process when the gas is :-
 (1) Expanded by adding more molecule to it.
 (2) Expanded by adding more heat to it.
 (3) Expanded against zero pressure.
 (4) Compressed by doing work on it.
72. Which one of the sols acts as lyophilic colloid ?
 (1) As_2S_3 (2) Gelatin
 (3) Au (4) $\text{Fe}(\text{OH})_3$
73. Among NO_3^- (I); AsO_3^{3-} (II), CO_3^{2-} (III) ClO_3^- (IV) and SO_3^{2-} (V). The multiplanar species are :-
 (1) II, IV, V (2) III, IV
 (3) I, II, V (4) II, III, IV
74. Which of the following show optical isomersim :-

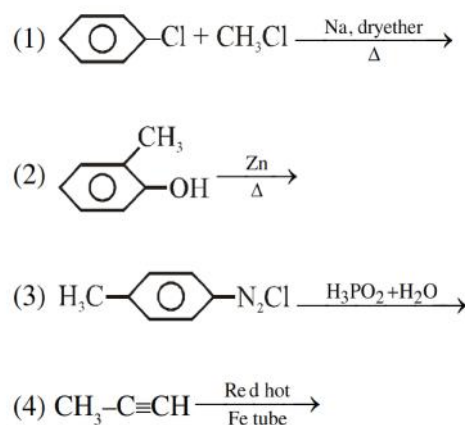


75. Which of following gives Fehling solution test ?
 (1)  (2) 
 (3)  (4) 
76. In the following structure, the oxidation number on carbon will be : $\text{H}-\text{N}\equiv\text{C}$:-
 (1) 0 (2) -2
 (3) +2 (4) +1 or -1
77. In which of the following reaction conc. H_2SO_4 is acts as an oxidising agent?
 (1) $\text{ZnO} + 2\text{H}_2\text{SO}_4 \longrightarrow \text{ZnSO}_4 + \text{H}_2\text{O}$
 (2) $2\text{HI} + \text{H}_2\text{SO}_4 \longrightarrow \text{I}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$
 (3) $\text{CuF}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{CuSO}_4 + 2\text{HF}$
 (4) $\text{Ba}(\text{OH})_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{BaSO}_4 + 2\text{H}_2\text{O}$

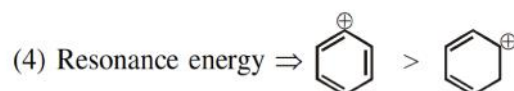
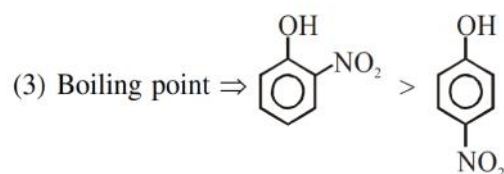
78. A solution of sodium in liquid ammonia is blue due to the presence of :-
 (1) Sodium atoms
 (2) Ammonium ions
 (3) Solvated sodium ion
 (4) Solvated electron



prepared by except :-



80. Which of following is correct ?
 (1) Boiling point $\Rightarrow$ n-pentane > diethyl ether
 (2) Solubility in water $\Rightarrow$ n-Butyl alcohol > isobutyl alcohol



81. A crystal is formed by two elements X and Y in cubic structure. X atoms are at the corners of a cube while Y atoms are at the face centre. The formula of the compound will be :

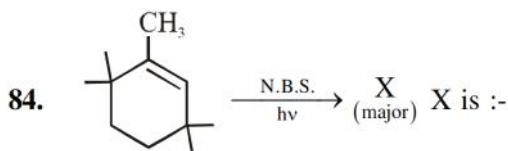
- (1) XY (2) XY₂
(3) X₂Y₃ (4) XY₃

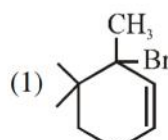
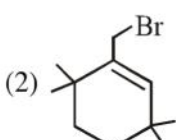
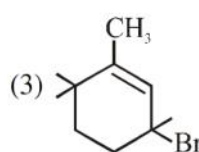
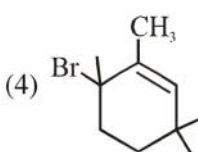
82. Main reason for brown ring test of NO₃⁻ is :-

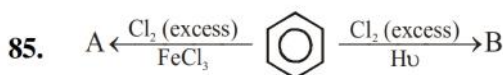
- (1) Negligible tendency of FeSO₄ to form complex
(2) Reducing power of FeSO₄ to convert NO₃⁻ to NO
(3) Decomposition of FeSO₄ to give SO₂ gas
(4) Oxidising property of FeSO₄

83. Chloroplatinic acid is :

- (1) monobasic (2) dibasic
(3) tribasic (4) tetrabasic



- (1)  (2) 
(3)  (4) 



What are A & B respectively :-

- (1) BHC and HCB
(2) HCB and BHC
(3) HCB and HCB
(4) BHC and BHC

86. In a mixture of A and B components show positive deviation when :-

- (1) A – B interaction is stronger than A – A and B – B interaction
(2) A – B interaction is weaker than A – A and B – B interaction
(3) $\Delta V_{\text{mix}} < 0$, $\Delta S_{\text{mix}} > 0$
(4) $\Delta V_{\text{mix}} = 0$, $\Delta S_{\text{mix}} > 0$

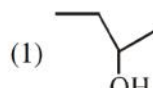
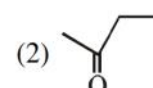
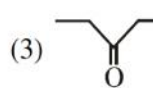
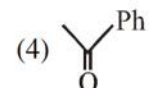
87. A black sulphide ppt when treated with ozone becomes white, the white compound is :-

- (1) ZnSO₄ (2) CaSO₄
(3) BaSO₄ (4) PbSO₄

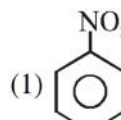
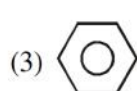
88. Which salt does not causes hardness in water ?

- (1) Mg(HCO₃)₂ (2) CaCl₂
(3) MgSO₄ (4) MgCO₃

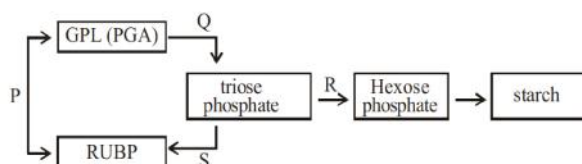
89. Which of the following does not show haloform reaction :-

- (1)  (2) 
(3)  (4) 

90. Which of following is correctly match :-

- (1)  $\Rightarrow$ oil of mirbane
(2) CH₃-S-C $\equiv$ N $\Rightarrow$ Mustard oil fragrance
(3)  $\Rightarrow$ Hinsberg reagent
(4) R-N₂⁺ Cl⁻ $\Rightarrow$ Diazo coupling

91. Protoxylem is towards periphery and metaxylem towards pith. Xylem is said to be :
 (1) Endarch (2) Polyarch
 (3) Misarch (4) Exarch
92. Heterosporous pteridophyte are :-
 (1) Lycopodium and pteris
 (2) Selaginella and psilotum
 (3) Selaginella and salvinia
 (4) Dryopteris and Adiantum
93. Non membranous organelle is :-
 (1) Chloroplast
 (2) Nucleolus
 (3) Centriole
 (4) Both (2) and (3)
94. The diagram represent the calvin cycle :-



at which stage CO_2 incorporated ?

- (1) P (2) Q (3) R (4) S
95. In which the following important intrinsic mechanisms provide auto regulation of glomerular filtration :-
 (1) Myogenic mechanism
 (2) Cardiac mechanism
 (3) Tubular secretion
 (4) Krebs Henseleit cycle
96. Post pollination development of pollen grain is known as :-
 (1) In-situ
 (2) Ex-situ
 (3) In-vitro
 (4) Precocious
97. All the following are uses of amniocentesis but one is misuse. Which one is misuse ?
 (1) It is done to examine the sex of the foetus leading to increasing female foeticides.
 (2) It is used to study metabolic defects of foetus like PKU (phenyl ketonuria)
 (3) It is used to defect foetal abnormalities.
 (4) It is used to detect genetic abnormalities.

98. Which of the following is best vector for plant and animal cells respectively :-
 (1) *Bacillus thuringiensis*, Rosie
 (2) *Pseudomonas putida*, Microinjection
 (3) Gene gun, *Agrobacterium tumifaciens*
 (4) *Agrobacterium tumifaciens*, Retroviruses
99. In his experiment S.L Miller observed the formation of :-
 (1) Proteins
 (2) Simple Amino Acids
 (3) Lipids
 (4) Pigments, N_2 Bases
100. If biotic potential of a species is high than :
 (1) Number of species in area is increase
 (2) Species diversity increase in area
 (3) Number of individuals of particular species are increase
 (4) None of these
101. Ovary is one chambered but it becomes two chambered due to formation of false septum in:
 (1) Brinjal and lemon
 (2) Mustard and capsella
 (3) Dianthus and primrose
 (4) Sunflower and marigold
102. Read the following statement. Find out how many statements are correct :-
 (A) The leaves in Gymnosperms are well adapted to with stand extremes of temperature, humidity and wind.
 (B) Gymnosperms are homosporous
 (C) Unlike Bryophyta and pteridophyta in gymnosperms the male and female gametophyte do not have an independent free-living existence.
 (D) In cycas male cone and megasporophyll are born on different trees.
 (1) Three (2) Four (3) One (4) Two
103. Which of the following statement regarding chromosome is not correct ?
 (1) At the time of cell division the chromatin material get condensed to form chromosome.
 (2) At metaphase stage each chromosome is consist of two cylindrical structure is called chromatids.
 (3) Kinetochore constitute actual site of attachment of spindle fiber to chromosome during cell division
 (4) Telomere promote fusion of one chromosome to other chromosome.

- 104.** If pyruvic acid accumulated in cell then which regulation required :-
- (1) Positive regulation By phasphofructokinase
 - (2) Allosteric modulation by pyruvic kinase enzyme
 - (3) Triose phosphate dehydrogenase enzyme inhibition.
 - (4) Link reaction acceleration
- 105.** Identify the correct and wrong statements :-
- (a) Human skeleton has 206 bones
 - (b) There are 12 pairs of ribs in human
 - (c) Sternum is present on the dorsal side of the body.
 - (d) All mammals (except of few) have 8 cervical vertebrae.
 - (e) Ribs are bicephalic
- | correct | wrong |
|-------------|-------|
| (1) a, b, c | d, e |
| (2) a, b, d | c, e |
| (3) a, b, e | c, d |
| (4) a, c, d | b, e |
- 106.** Polar nuclei are situated in the central cell :-
- (1) Below the egg apparatus
 - (2) Above the egg apparatus
 - (3) Below the antipodals
 - (4) Near to micropyle
- 107.** Which of the following is incorrectly matched ?
- (1) Non-medicated IUD's – Lippes loop
 - (2) Copper releasing IUD – LNG - 20
 - (3) Hormone releasing IUD – Progestasert
 - (4) Saheli – Once a week contraceptive pill
- 108.** Recombinant DNA technology involves several steps in specific sequence. What is the correct sequence for r-DNA technology ?
- a → Down stream processing.
 - b → Cutting of DNA at specific location.
 - c → Obtaining the foreign gene product.
 - d → Isolation of the genetic material.
 - e → Insertion of r-DNA into the host cell.
- (1) b → d → e → a → c
 - (2) d → b → e → c → a
 - (3) d → e → d → a → c
 - (4) e → d → a → c → b

- 109.** Select the incorrect statement :-
- (1) Lichen can be used as industrial pollution indicators.
 - (2) Evolution is a stochastic process based on chance event in nature and chance mutation in the organism.
 - (3) Evolution is a directed process in the sense of determinism.
 - (4) Excess use of herbicides, pesticides, etc has only resulted in selection of resistant varieties in a much lesser time scale.
- 110.** How many statements are correct :
- (a) Predators also help in maintaining species diversity in a community
 - (b) Decomposition is largely an oxygen requiring process
 - (c) Number of trophic levels in GFC is never restricted
 - (d) Some of organisms of DFC are prey to GFC animals
- (1) Three
 - (2) One
 - (3) Two
 - (4) All of the above
- 111.** Which one of the following is enucleated dead cell ?
- (1) Companion cell
 - (2) Sieve tube
 - (3) Guard cell
 - (4) Vessel
- 112.** Who discovered viroids :-
- (1) Stanley B prusiner
 - (2) T.O. Diener
 - (3) D.J. Ivanowsky
 - (4) W.M. Stanley
- 113.** Which one is not found in matrix of mitochondria?
- (1) 70-S Ribosome
 - (2) Kreb's cycle enzyme
 - (3) Single strand DNA molecule
 - (4) RNA molecule
- 114.** ATP is injected in cyanide poisoning because it is :-
- (1) Neccessary for cellular function
 - (2) Neccessary for $\text{Na}^+ - \text{K}^+$ pump
 - (3) $\text{Na}^+ - \text{K}^+$ pump operates at the cell membraneous
 - (4) ATP breakdown cyanide.

115. During forensic investigation, a pelvic girdle was found, is of female. How does they concluded that it was of female not male ?

- (1) Pelvic girdle diameter is less in female's than male's
- (2) Pelvic diameter is more in female's than male's
- (3) Female pelvis has one obturator foramen
- (4) Pubic symphysis is not fused in females

116. Which of the following is formed as a result of triple fusion ?

- (1) Fruit
- (2) Seed
- (3) Perisperm
- (4) Endosperm

117. Which one of the following is incorrect for garden pea :-

	Character	Genotype	Phenotype
(1)	Flower position	Homozygous or heterozygous	Terminal
(2)	Seed colour	Homozygous or heterozygous	Yellow
(3)	Pod colour	Homozygous or heterozygous	Green
(4)	Pod shape	Only Homozygous	Constricted

118. What out come is expected in offsprings if father is colourblind and normal mother's father was colourblind ?

- (1) 50% daughter colourblind
- (2) All daughters colourblind
- (3) All sons are colourblind
- (4) All sons are normal

119. Neopilina is a connecting link between :-

- (1) Annelida and mollusca
- (2) Annelida and Arthropoda
- (3) Amphibia and reptilia
- (4) Arthropoda and Mollusca

120. Fish die in polluted water by sewage due to :

- (1) Pathogen
- (2) Biomagnification
- (3) Foul smell
- (4) Reduction in O₂

121. Select the mismatch from the following :

- (1) Cruciferae – Tetradynamous
- (2) Solanaceae – Perianth
- (3) Liliaceae – Epiphyllous
- (4) Fabaceae – Vexillary

122. The correct classification of Trygon is :-

- (1) Chordata → Protochordata → Vertebrata → Pisces → Chondrichthyes
- (2) Chordata → Ganathostomata → Pisces → Osteichthyes
- (3) Choradata → Ganathostomata → Pisces → Chondrichthyes
- (4) Choradata → Agnatha → Cyclostomata

123. Match the column-I and column-II and choose the correct option :-

Column-I		Column-II	
(A)	Synapsis	(i)	Anaphase-II
(B)	Synthesis of RNA and protein	(ii)	Zygotene
(C)	Action of enzyme recombinase	(iii)	G ₂ -Phase
(D)	Centromere do not seprate but chromatids move towards opposite pole.	(iv)	Anaphase-I
		(v)	Pachytene

- (1) A-iii, B-i, C-v, D-ii
- (2) A-ii, B-iii, C-v, D-iv
- (3) A-ii, B-i, C-iv, D-v
- (4) A-v, B-iii, C-iv, D-i

124. Plant hormone associated with climacteric respiration in :-

- (1) Auxin
- (2) Cytokinin
- (3) Ethylene
- (4) Gibberellin

125. Typical cervical vertebra has :-

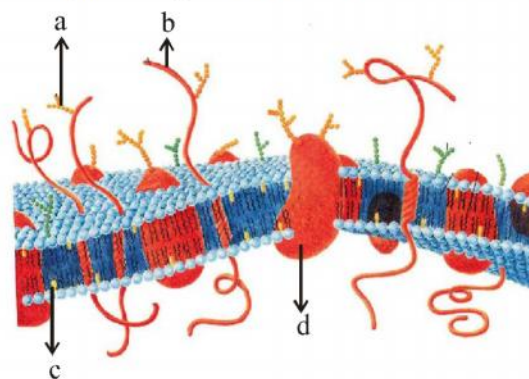
- (1) Spinal foramen
- (2) Transverse process
- (3) Foramen Tranversarium
- (4) Lamina

126. If the embryosac cell of an angiosperm has 12 chromosome then in the microspore mother cell will be :-

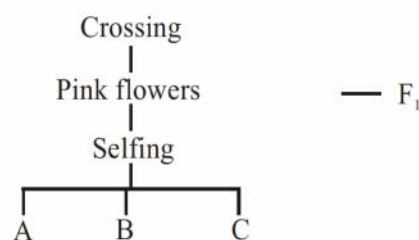
- (1) 12
- (2) 24
- (3) 36
- (4) 48

127. The phenotype is affected when :-
 (1) Modified allele produces normal enzyme
 (2) Modified allele produces non functional enzyme
 (3) Unmodified allele produces normal enzyme
 (4) Both 2 and 3
128. 'Restriction' in restriction enzyme refers to :-
 (1) Cleaving of phosphodiester bond in DNA by enzyme
 (2) Cutting of DNA at specific position only
 (3) Prevention of multiplication of bacteriophages in bacteria
 (4) All of the above
129. Antibodies which can pass through placenta, found abundantly in colostrum and involved in allergy respectively are
 (1) Ig E, Ig A and Ig G (2) Ig G, Ig E and Ig A
 (3) Ig A, Ig E and Ig G (4) Ig G, Ig A and Ig E
130. Read statement (a-d) regarding population interaction :
 (a) Human liver fluke has two intermediate host for survival
 (b) Cuscuta is a parasitic plant lost its chlorophyll during evolution
 (c) Female mosquito is considered as parasite
 (d) Morphological and anatomical features are highly complex of endoparasites
 Which of following statement is incorrect :
 (1) a, b and c (2) a, c and d
 (3) b, c and d (4) c and d only
131. Which is not a stem modification :
 (1) Tuber of potato
 (2) Offset of pistia
 (3) Leaf bladder of utricularia
 (4) Corm of colocasia
132. Which one of the following phyla is correctly matched with its two general characteristics ?
 (1) Mollusca - Normally oviparous and development through a trochophore or veliger larva
 (2) Arthropoda- Body divided into head thorax and abdomen and respiration by tracheae
 (3) Chordata - Notochord at some stage and separate anal and urinary opening to the outside
 (4) Echinodermata - Pentamerous radial symmetry and mostly internal fertilization

133. Recognise the figure and find out the correct matching:-



- (1) a-Protein, b-Integral protein, c-Sugar, d-Cholesterol
 (2) a-Sugar, b-Protein, c-Cholesterol, d-Integral protein
 (3) a-Cholesterol, b-Sugar, c-Protein, d-Integral protein
 (4) a-Integral protein, b-Sugar, c-Cholesterol, d-Protein
134. An enzyme that can stimulate germination of barley seeds is :-
 (1) Lipase (2) Protease
 (3) Invertase (4) α -amylase
135. Cerebral hemispheres are connected by a tract of nerve fibres called as :-
 (1) Corpus albicans (2) Corpus callosum
 (3) Thalamus (4) 2 and 3 both
136. When pollen grain of a flower pollinate the stigma of another flower of the same plant, it is called:-
 (1) Xenogame (2) Herkogamy
 (3) Geitonogamy (4) Dichogamy
137. A cross is being given here :-
 Red flowers White flowers — P



If B stands for pink flower then

- (a) What is the name of plant ?
 (b) What % of F_2 progeny will produce A ?
- (1) (a) Garden pea (b) 25%
 (2) (a) Snapdragon (b) 50%
 (3) (a) Snapdragon (b) 25%
 (4) (a) Garden pea (b) 50%

138. Name of microbe used for production of Swiss cheese :-
 (1) *Sacchromyces cerevisiae*
 (2) *Propionibacterium sharmanii*
 (3) *Monascus purpureus*
 (4) *Lactobacillus*
139. Metastasis can be defined as
 (1) the spread of cancer from one person to another
 (2) the spreading process of sexually transmitted diseases.
 (3) the spreading process of communicable diseases from one organ to another.
 (4) the spreading process of cancer from one organ to another through body fluids.
140. Slash and burn agriculture commonly called as jhum cultivation in North-Eastern state of India related to :
 (1) Deforestation (2) Afforestation
 (3) Crop rotation (4) Dry farming
141. Term "Taxonomy" was given by :-
 (1) Carolus Linnaeus (2) Gaspard Bauhin
 (3) A.P. Candolle (4) J. Huxley
142. Cylindrical bodies, worm like marine animals belong to the phylum :-
 (1) Echinodermata
 (2) Mollusca
 (3) Coelenterata
 (4) Hemichordata
143. Which of the following is a rapid type of absorption :-
 (1) Passive absorption (2) Active absorption
 (3) Salt absorption (4) Root absorption
144. A doctor advises a patient to include yellow fruits carrots and butter in his diet. What deficiency disease do you think the patient is suffering from :-
 (1) Colour blindness
 (2) Scurvy disease
 (3) Night blindness
 (4) Marasmus disease
145. Forebrain does not include :-
 (1) Cerebrum
 (2) Thalamus
 (3) Hypothalamus
 (4) Corpora quadrigemina

146. A aquatic plant, water lily, is pollinated by :-
 (1) water (2) air
 (3) insect (4) birds
147. Which one of the following is not a mendelian disorder?
 (1) Haemophilia (2) Sickle cell anaemia
 (3) Phenylketonuria (4) Down's syndrome
148. In cells of superfemale with 47 chromosomes (44 + XXX), visible barr bodies are :-
 (1) 1 (2) 0
 (3) 2 (4) 3
149. Mark the incorrect combination
 (1) Aquaculture- crab, lobsters and edible oyster
 (2) Fresh water fishes- Catla, rohu, common carp and pomfrets
 (3) Marine fishes- Hilsa, Sardine, Mackerel
 (4) MOET- Cattle, sheep rabbit, buffaloes and mares
150. Logistic growth and exponential growth formula is :
 (1) $\frac{dN}{dt} = rN$; $\frac{dN}{dt} = rN \left(\frac{N-K}{N} \right)$
 (2) $\frac{dN}{dt} = rN \left(\frac{K-N}{K} \right)$; $\frac{dN}{dt} = rN$
 (3) $\frac{dN}{dt} = rN$; $\frac{dN}{dt} = rN$
 (4) $\frac{dN}{dt} = rN$; $\frac{dN}{dt} = rN \frac{(K-N)}{N}$
151. In *Magnifera indica*, *indica* represents :-
 (1) Generic name (2) Name of author
 (3) Specific epithet (4) Subspecific name
152. Match the columns I, II, and III and choose the correct combination from the options given :-

	Column-I		Column-II		Column-III
A.	Testes	(i)	2 nd -6 th Segments	(P)	1 Pair
B.	Ovaries	(ii)	4 th - 6 th Segments	(Q)	1 Pair
C.	Spermathecae	(iii)	6 th - 7 th Segments	(R)	Double
D.	Mush room gland	(iv)	6 th - Segment	(S)	Single

- (1) A-(ii)-R, B-(i)-Q, C-(iii)-P D-(iv)-P
 (2) A-(ii)-P, B-(i)-Q, C-(iv)-P, D-(iii)-S
 (3) A-(i)-Q, B-(ii)-P, C-(iv)-R, D-(iii)-S
 (4) A-(iii)-P, B-(iv)-R, C-(i)-Q, D-(ii)-S

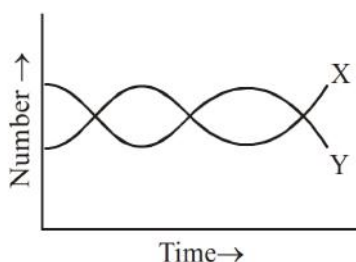
153. Cell A and B are adjacent plant cells. In cell A, $\psi_s = -20$ bars and $\psi_p = 8$ bars. In cell B, $\psi_s = -12$ bars and $\psi_p = 2$ bars. Then :-
 (1) Water moves from cell A to cell B
 (2) There is no movement of water between cells A and cell B.
 (3) Water moves from cell B to cell A
 (4) Equal amount of water is simultaneously exchanged between cell A and cell B.
154. Which one of these statement about intrinsic factor is not true ?
 (1) It is a glycoprotein
 (2) It is secreted by gastric gland
 (3) It helps to prevent pernicious anaemia
 (4) It promotes absorption of vitamin B₁₂ in the intestine
155. Disease caused by hypo and hypersecretion of collip's hormone is :-
 (1) Tetany and Osteoporosis
 (2) Osteoporosis and Tetany
 (3) Tetany and Asphyxia
 (4) Asphyxia and Cretinism
156. The Graafian follicle ruptures to release _____ from the ovary by the process called ovulation.
 (1) Primary oocyte
 (2) Secondary oocyte after completing meiosis-II.
 (3) Secondary oocyte after completing meiosis-I & with the release of Ist polar body.
 (4) Mature ovum
157. Select incorrect statement regarding DNA replication ?
 (1) Leading strand is formed in 5' → 3' direction.
 (2) Okazaki fragments are formed in 5' → 3' direction
 (3) DNA polymerase catalyzes polymerisation in 5' → 3' direction
 (4) Continuous strand synthesizes from 5' → 3' and discontinuous strand from 3' → 5' direction
158. A gene showing codominance has :
 (1) both alleles independently expressed in the heterozygote
 (2) one allele dominant on the other
 (3) alleles tightly linked on the same chromosome
 (4) alleles that are recessive to each other

159. Which of the following species of plasmodium cause most serious and fatal malaria.
 (1) P.vivax (2) p. falciparum
 (3) P. malariae (4) P. ovale
160. In equation $\frac{dN}{dt} = rN \left[\frac{K - N}{K} \right]$; here K is defined by :
 (1) Maximum population density achievable under available resources
 (2) Carrying capacity of habitat
 (3) Both (1) and (2)
 (4) Maximum intrinsic rate of growth
161. Which of the following statement is incorrect ?
 (1) Fungi prefer to grow in cold and humid places.
 (2) Karyogamy in fungi occurs after plasmogamy
 (3) Yeast are used to make bread and beer.
 (4) Fungi bodies consist of long slender thread like structure called mycelium
162. Virus particle outside the host body is called
 (1) Virion (2) Prion
 (3) Viriods (4) Bacteriophage
163. Select the correct match :-
 A *Nitrosomonas* Nitrite to nitrate
 B *Thiobacillus* Denitrification
 C *Nastoc* Free-living nitrogen-fixer
 D *Azotobacter* Anaerobic nitrogen-fixer
 (1) A and B (2) C and D
 (3) B and C (4) B and D
164. Which type of white blood cells are concerned with the phagocytic in nature and maximum in number ?
 (1) Monocytes (2) Eosinophils
 (3) Basophils (4) Neutrophils
165. Match the correct sets :-

	Endocrine	Hormone	Deficiency
(a)	Adenohypophysis	Vasopressin	Diabetes Mellitus
(b)	Parathroid gland	Parathormone	Tetany
(c)	Pancreas	Insulin	Diabetes Mellitus
(d)	Thyroid gland	Calcitonin	Acromegaly

- (1) a, d (2) b, d
 (3) b, c (4) a, b

166. Signals from fully developed foetus and placenta ultimately leads to parturition which requires the release of :
 (1) Estrogen from placenta
 (2) Oxytocin from foetal pituitary
 (3) Oxytocin from maternal pituitary
 (4) Relaxin from leydigs cells
167. If the base sequence of a codon in mRNA is 5'-AUG-3', the sequence of tRNA pairing with it must be :-
 (1) 5'-AUG-3' (2) 5'-UAC-3'
 (3) 5'-CAU-3' (4) 5'-GAU-3'
168. C-DNA molecule synthesized against m-RNA using reverse transcriptase has a length which is :-
 (1) Lesser than in vivo gene
 (2) Lesser than mRNA
 (3) Same as in vivo gene
 (4) Greater Than mRNA
169. A person is having dry, scaly lesions on various parts of the body such as skin, nails and Scalp. These lesions are accompanied by intense itching. This disease may be caused by
 (1) Microsporum and Entamoeba
 (2) Trichophyton and microsporum
 (3) Epidermophyton and Plasmodium
 (4) Entamoeba and Plasmodium
170. What type of interaction shows in this graph between two organism X and Y :



- (1) Mutualism (2) Commensalism
 (3) Parasitism (4) Predation
171. Read the following characters find out which one is not related with Gymnosperm :-
 (1) Association with N_2 fixing cyanobacteria
 (2) Mycorrhiza
 (3) More than one archegonia
 (4) Independent free living gametophyte

172. Read the following statement carefully (A-D) :-
 (a) Anything less than a complete structure of a cell does not ensure independent living.
 (b) Ribosome are membrane bound organelles found in prokaryotic and Eukaryotic cell.
 (c) Cytoplasm is the main arena (Zone) of cellular activities in both plant and animals cell.
 (d) Movement of water by diffusion is called osmosis.
 How many statement are correct ?
 (1) One (2) Two (3) Three (4) Four
173. The rate of photosynthesis does not depend upon :-
 (1) Light duration (2) Light intensity
 (3) Light quality (colour) (4) Temperature
174. Which blood vessels contains the more amount of urea ?
 (1) Hepatic vein (2) Renal vein
 (3) Hepatic Artery (4) Renal Artery
175. Hormone that initiates ejection of milk and promote production of antibodies to provide humoral immunity :-
 (1) Oxytocin and Thymosins
 (2) Prolactin and Thymosins
 (3) Oxytocin and Cortisol
 (4) Follicle stimulating hormone and Cortisol
176. Identify the hormones that are secreted is large amount prior to ovulation.
 A. LH B. FSH
 C. Progesterone D. Estrogen
 (1) Only A (2) Only A and D
 (3) A, B & D (4) A, B, C, D

A	B
Occur in pairs	Occur in pairs
Segregate at the time of gamete formation such that only one of each pair is transmitted to a gamete	Segregate at gamete formation and only one of each pair is transmitted
Independent pairs segregate independently of each other	One pair segregates independently of another pair

Select out the incorrect conclusion :-

- (1) All the statements of column A are true for chromosome
 (2) All the statements of column B are always true for chromosome
 (3) All these statements of column A are always true for Gene
 (4) All the statements of column B are always true for Gene

178. Which of the following statement is false :-

- (1) The 1st cellular form of life did not possibly originate till About 2000 mya.
- (2) The geological history of earth closely correlates with biological history of earth.
- (3) According special creation theory earth is about 4.5 billion year old.
- (4) The universe is very old-almost 20 billion years old.

179. How many statements are correct :

- (a) Domestic sewage contains biodegradable organic matter.
- (b) A scrubber remove gases like sulphur dioxide
- (c) Electrostatic precipitates can remove 99% particulate matter present in exhaust from thermal power plant.
- (d) Particulate size 2.5 μm or less in diameter are not harmful for human health

- | | |
|----------|-----------|
| (1) Two | (2) One |
| (3) Four | (4) Three |

180. Desert, Grassland, Tropical forest, Temperate forest, Coniferous forest and Tundra.

Which of the following biome research area of Ramdeo Mishra ?

- (1) Tundra and Temperate forest
- (2) Desert and Grassland
- (3) Tropical forest and Desert
- (4) Tropical forest and Grassland